

Name:

Roll No.

Section:

Midsem Exam Room No. :

Seat No.:

MS101 – Makerspace
2022-23/I (Autumn Semester)

Dec 20, 2022 (Tue)

Midsemester Examination (EE)

Time: 40 min

Marks: 30

1. This **Question-cum-Answer Booklet** has 4 pages.
2. Write your **answers only in the space provided for answers**. Answers written at any other place will not be checked.
3. No explanations/clarifications will be given to any of the questions.
4. You must show steps of your answers/calculations inside the booklet in the space provided. You may use the additional “Supplementary Sheet” given to you for any extra work. You have to compulsorily attach the Supplementary Sheet along with the Booklet.
5. No negative marks for wrong answers.

1. Out of the following devices, **choose ALL the ones** which are single-port devices
- a. Inductor
 - b. Transformer
 - c. Diode
 - d. Bipolar Junction Transistor
 - e. Capacitor

Marks: 1

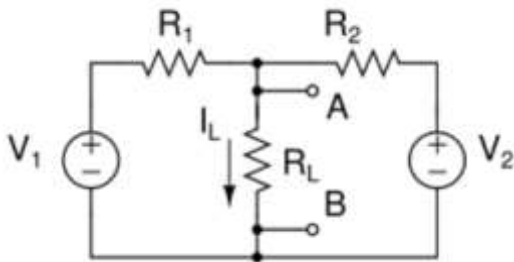
Answer(s):

a c e

2. In the resistive network shown below, $V_1 = 10\text{ V}$, $V_2 = 15\text{ V}$, $R_1 = 2\text{ k}\Omega$ and $R_2 = 3\text{ k}\Omega$.

Apply Thevenin's theorem across nodes A and B, and calculate current I_L in **mA** for $R_L = 2.8\text{ k}\Omega$ and $R_L = 1.3\text{ k}\Omega$.

Show your steps in the space below.



Marks: 4 (=1+1+1+1)

Answers:

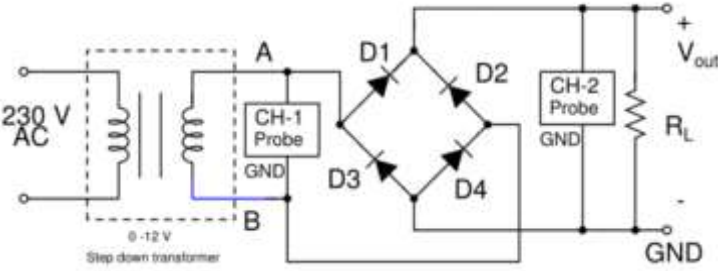
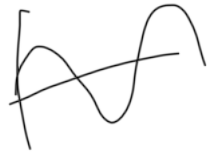
 $V_{Th} =$ $R_{Th} =$
 $I_L \text{ (for } R_L = 2.8\text{ k}\Omega\text{)}$
 $=$ mA

 $I_L \text{ (for } R_L = 1.3\text{ k}\Omega\text{)}$
 $=$ mA

Write Steps below:

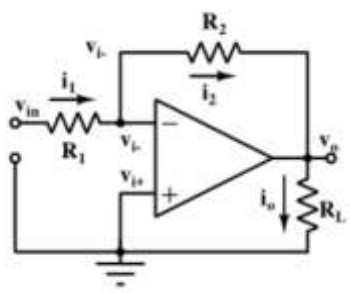
3. The circuit diagram of the bridge rectifier used in Expt 2 is shown

Answers

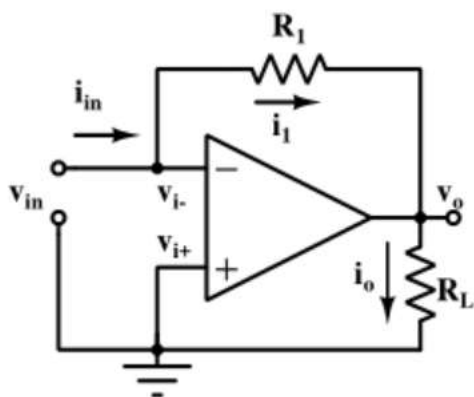
below. One MS101 student wanted to see both V_{AB} and V_{out} of the bridge rectifier circuit on the DSO. Hence, he connected CH-1 of the DSO through the DSO probe to the transformer secondary terminals (see the figure). He then connected CH-2 of the DSO through another DSO probe to V_{out} (see the figure).  a) Sketch the V_{AB} and V_{out} waveforms that he would see on CH-1 and CH-2 channels (with the connections as described). Marks: 2x2 = 4	V_{AB} waveform  a) V_{out} waveform 
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4. Choose the correct option by indicating the letter corresponding to the correct statement. An ideal op amp has finite output voltage with zero differential voltage because it has: (A) infinite differential voltage gain. (B) infinite input resistances for the two inputs. (C) infinite common-mode rejection ratio. (D) dual supply voltages.	Marks: 1 Answer: B
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For the following two questions, state whether the statement is ‘True’ or ‘False’.	Marks: 2x1 = 2
5. Noise is always a random waveform.	Answer: F
6. The two supply voltages in a dual-supply amplifier have to be equal.	Answer: F

7. The circuit in the figure has a load resistance R_L connected between the amplifier output and ground, and the output current i_o flows into this load. $R_1 = 15\text{ k}\Omega$, $R_2 = 150\text{ k}\Omega$. $R_L = 1\text{ k}\Omega$ 	Marks: 4x1 = 4 (A) For $v_{in} = +120\text{ mV}$, $i_2 = \underline{\hspace{2cm}}$ mA. (B) For $v_{in} = -120\text{ mV}$, $v_o = \underline{\hspace{2cm}}$ mV. (C) Current gain $A_i = \underline{\hspace{2cm}}$. (D) Power gain $A_p = \underline{\hspace{2cm}}$.
8. The circuit in the figure has a load resistance R_L	Marks: 2x1 = 2

connected between the amplifier output and ground. The output current i_o flows into this load. $R_1 = R_L = 800\ \Omega$.

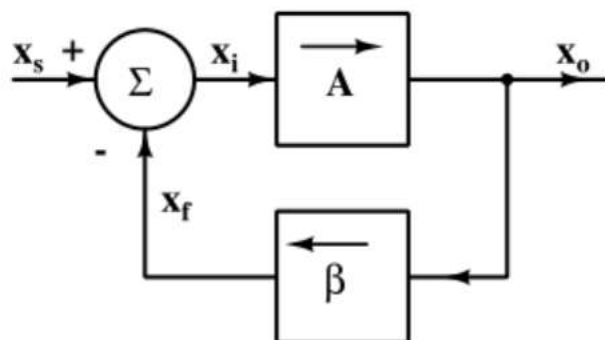


For this circuit,

(A) $R_{in} = \underline{\hspace{2cm}}$ k Ω .

(B) For $i_{in} = 2\text{ mA}$, $i_o = \underline{\hspace{2cm}}$ mA.

9. In the general feedback structure in the figure,



Marks: 4 (= 1+ 1 +1 +1)

(A) $x_o =$
(write x_o in terms of x_s , A and β)

(B) Loop-gain =

(C) Closed-loop gain $A_f =$

(D) For $A \gg 1/\beta$, $A_f =$

For the following two questions, choose the correct option by indicating the letter corresponding to the correct statement.

Marks : 1

10. What will happen if the power terminals of a DC motor are reversed?

Answer:

- (A) Very high current will flow through the motor and possibly damage the motor
- (B) DC motor has 'reverse polarity protection' and hence no damage will be done, but the motor will not start.
- (C) The motor will rotate in the opposite direction.
- (D) The motor will stall (will not be able to move).

11. While representing the transistor ac amplifier operation on the VCE-IC graph, the operating point moves along the.....

Marks : 1

Answer:

- (A) ' $V_{CE}=\text{constant}$ ' line
- (B) Saturation region line
- (C) ' $I_C=\text{constant}$ ' line
- (D) Load line

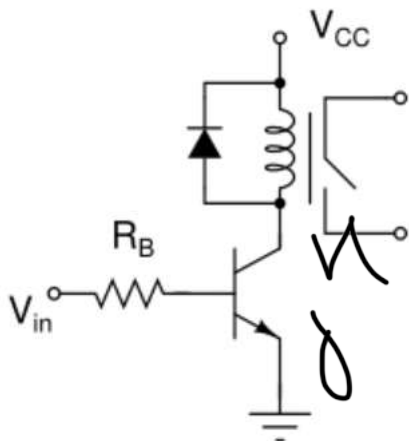
12. A BJT based relay circuit is shown below

Marks: 3

which uses an *npn* transistor. The relay requires a minimum of 20 mA to operate.

(a) Calculate the minimum value of V_{in} required to operate the relay.

Given: $V_{CC} = 12\text{ V}$, $R_B = 10\text{ k}\Omega$, $V_{BE} = 0.7\text{ V}$,
 $\beta = 50$, $R_{Coil} = 400\text{ }\Omega$, $V_{CEsat} = 0.2\text{ V}$
 (R_{Coil} is the resistance of the relay coil)



Space for steps and answer

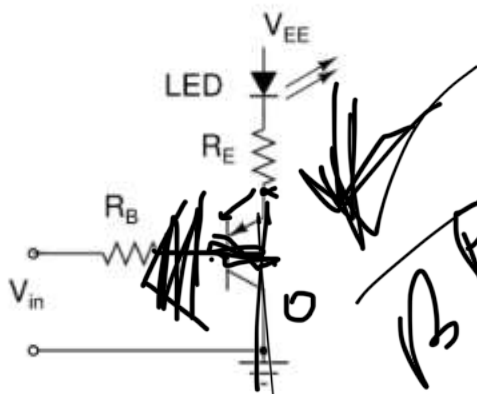
(a) Minimum value of V_{in} required = _____ V

13. The switching circuit shown below uses a *pnp* transistor. Calculate the current through the LED in mA when:

i) $V_{in} = 0\text{ V}$ and ii) $V_{in} = V_{EE}$.

You may neglect the base current.

Given: $V_{EE} = 12\text{ V}$, $V_{BE} = -0.7\text{ V}$, $R_B = 20\text{ k}\Omega$, $R_E = 2\text{ k}\Omega$, $V_{LED} = 2\text{ V}$
 (V_{LED} is the voltage drop across the LED when it is ON)



Marks: 3

Show steps of your calculation:

Answers:

(i)

(ii)